

AI Assisted Coding

Assignment-16.3

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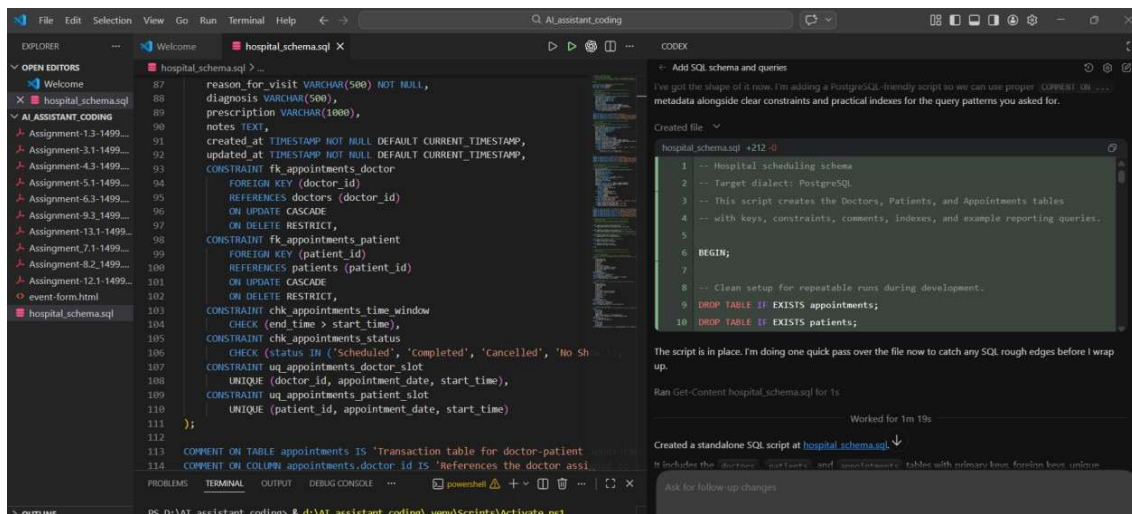
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Batch-08

Lab 16 – Database Design and Queries: Schema Design and SQL Generation

Task 1 – Hospital Management Database

Prompt: “Create SQL schema (Doctors, Patients, Appointments) with keys & constraints. Add queries: doctor appointments, patient history, patient count per doctor. Include comments + indexing.”

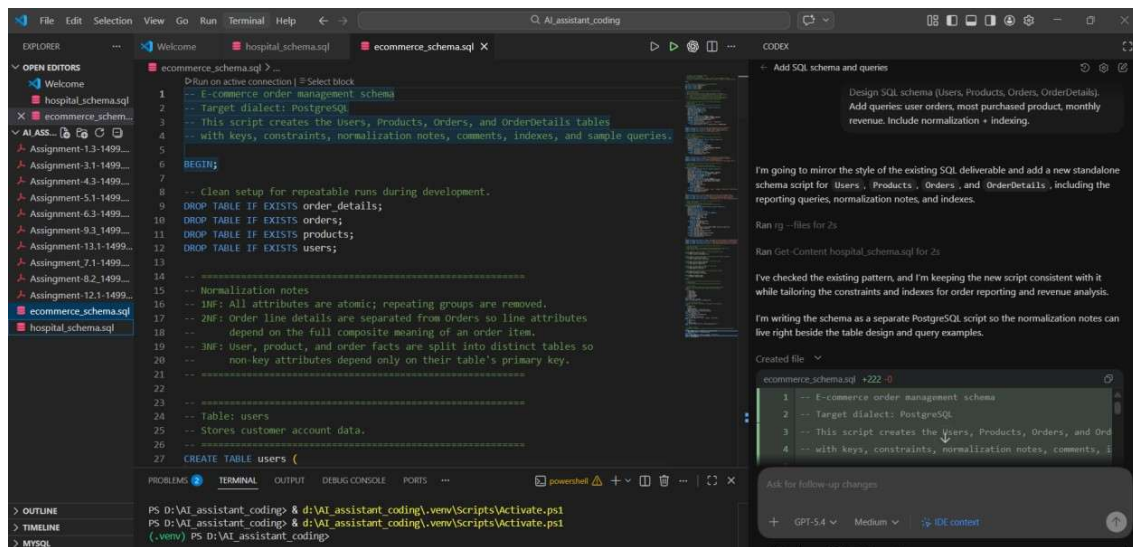


The screenshot shows a code editor with a file explorer on the left, a main editor window, and a right-hand panel. The file explorer shows a project named 'hospital_schema.sql' with files like 'hospital_schema.sql', 'AI ASSISTANT CODING', and various assignment files. The main editor window displays the SQL schema for the 'hospital_schema.sql' file. The schema includes tables for 'doctors', 'patients', and 'appointments'. The 'doctors' table has columns for 'reason_for_visit', 'diagnosis', 'prescription', 'notes', 'created_at', and 'updated_at'. The 'patients' table has columns for 'patient_id', 'patient_name', 'patient_age', 'patient_gender', 'patient_address', and 'patient_contact'. The 'appointments' table has columns for 'appointment_id', 'doctor_id', 'patient_id', 'appointment_date', 'appointment_time', 'appointment_status', and 'appointment_notes'. The schema includes foreign key constraints between 'doctors' and 'appointments', and between 'patients' and 'appointments'. It also includes check constraints for 'appointment_status' and 'appointment_time'. The right-hand panel shows a preview of the SQL schema and a list of queries: 'doctor appointments', 'patient history', and 'patient count per doctor'. It also includes a section for 'Include comments + indexing'.

```
1 -- Hospital scheduling schema
2 -- Target dialect: PostgreSQL
3 -- This script creates the Doctors, Patients, and Appointments tables
4 -- with keys, constraints, comments, indexes, and example reporting queries.
5
6 BEGIN;
7
8 -- Clean setup for repeatable runs during development.
9 DROP TABLE IF EXISTS appointments;
10 DROP TABLE IF EXISTS patients;
```

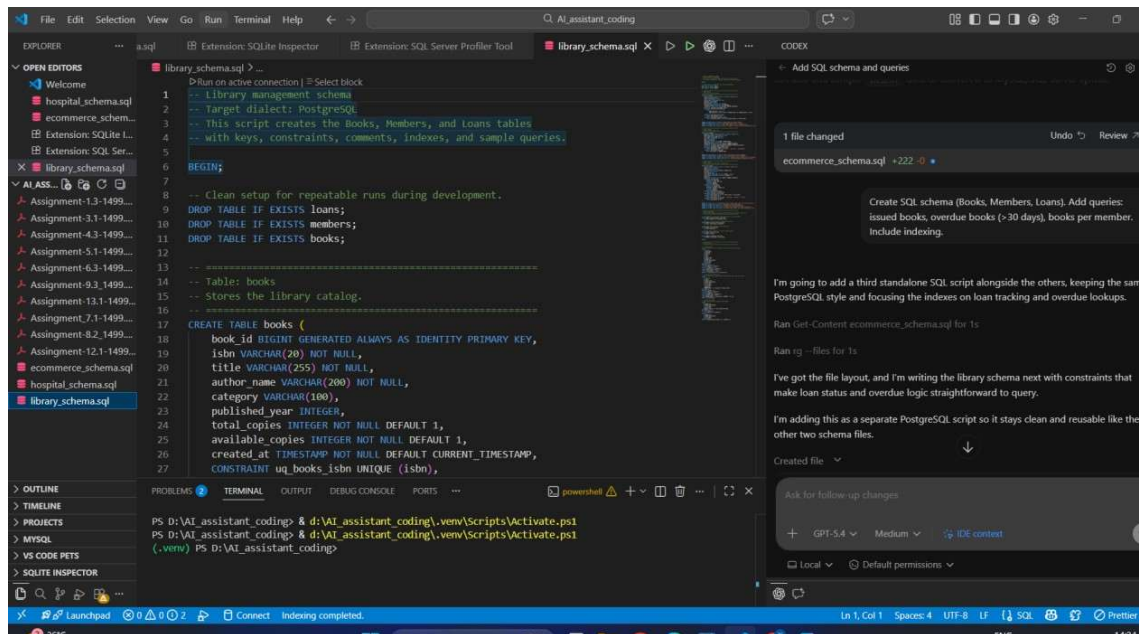
Task 2 – Real-Time Application: Online Shopping Database

Prompt: “Design SQL schema (Users, Products, Orders, OrderDetails). Add queries: user orders, most purchased product, monthly revenue. Include normalization + indexing.”



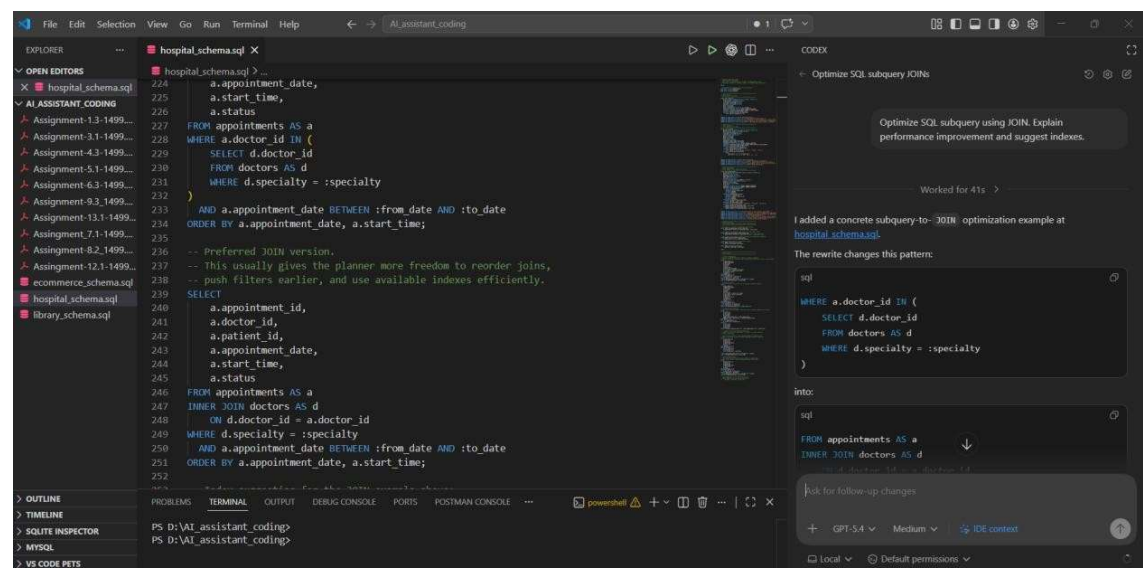
Task 3 – Library Database

Prompt: “Create SQL schema (Books, Members, Loans). Add queries: issued books, overdue books (>30 days), books per member. Include indexing.”



Task 4: Optimize Queries

Prompt: “Optimize SQL subquery using JOIN. Explain performance improvement and suggest indexes.”



Task 5: University Course Registration

Prompt: “Design SQL schema (Students, Courses, Faculty, Registrations). Add queries: students per course, faculty >2 courses, most registered course. Include optimization.”

